

AOS Assignment-1 Part-A

Bratin Mondal 21CS10016

Datta Ksheeraj 21CS30037

Building New Kernel

Step 1: Download the kernel source from the
www.kernel.org

Step 2: Extract the kernel source and in terminal navigate to the kernel source directory

1. `sudo apt-get install build-essential libncurses-dev bison flex libssl-dev`
2. `make menuconfig`

Step 3: In the menuconfig, select the required configuration and save the configuration

Specifically turn off the IPV6 support in the kernel configuration by navigating to the following path in the menuconfig

```
Networking support --->
Networking options --->
<*> The IPv6 protocol
```

To make KVM an inbuilt feature instead of a kernel module, navigate to the following path in the menuconfig

```
Virtualization --->
<*> Kernel-based Virtual Machine (KVM) support
```

Step 4: Save the configuration as .config file and exit the menuconfig

Step 5: Need to change some part of the .config file manually. The keys section in the .config file should be changed as follows

```
#
# Certificates for signature checking
#
CONFIG_MODULE_SIG_KEY="certs/signing_key.pem"
CONFIG_SYSTEM_TRUSTED_KEYRING=y
CONFIG_SYSTEM_TRUSTED_KEYS=""
CONFIG_SYSTEM_EXTRA_CERTIFICATE=y
CONFIG_SYSTEM_EXTRA_CERTIFICATE_SIZE=4096
CONFIG_SECONDARY_TRUSTED_KEYRING=y
CONFIG_SYSTEM_BLACKLIST_KEYRING=y
CONFIG_SYSTEM_BLACKLIST_HASH_LIST=""
CONFIG_SYSTEM_REVOCATION_LIST=y
CONFIG_SYSTEM_REVOCATION_KEYS=""
# end of Certificates for signature checking
```

Step 6: Some Installation of required packages

```
sudo apt-get install libelf-dev
sudo apt install dwarves
```

Step 7: Build the kernel

```
make -j4 or make -j$(nproc)
```

Step 8: Install the kernel

```
sudo make modules_install
sudo make install
```

Step 9 : Update Grub and Reboot

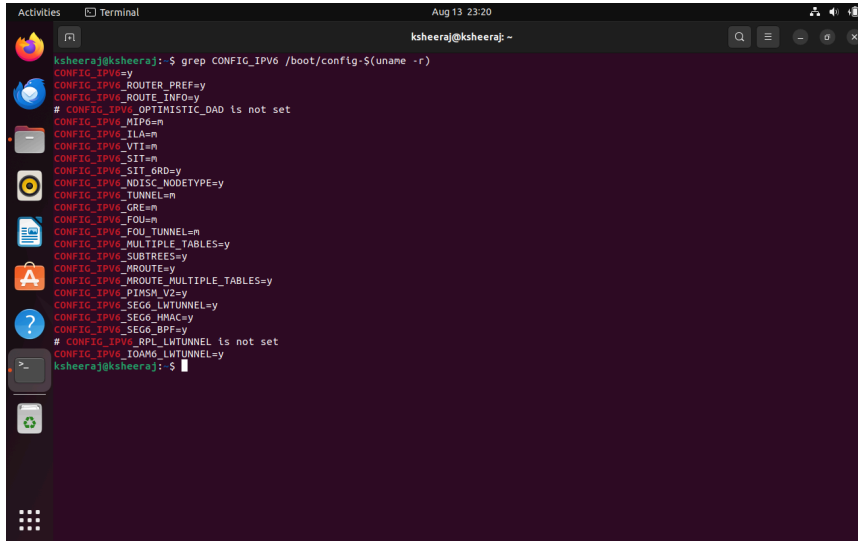
```
sudo update-grub
sudo reboot
```

We can choose to change the grub default or choose the required kernel by first changing the **GRUB_TIMEOUT** in **/etc/defaultgrub** from 0 to 5/10 seconds

Changes Observed:

1. IPV6

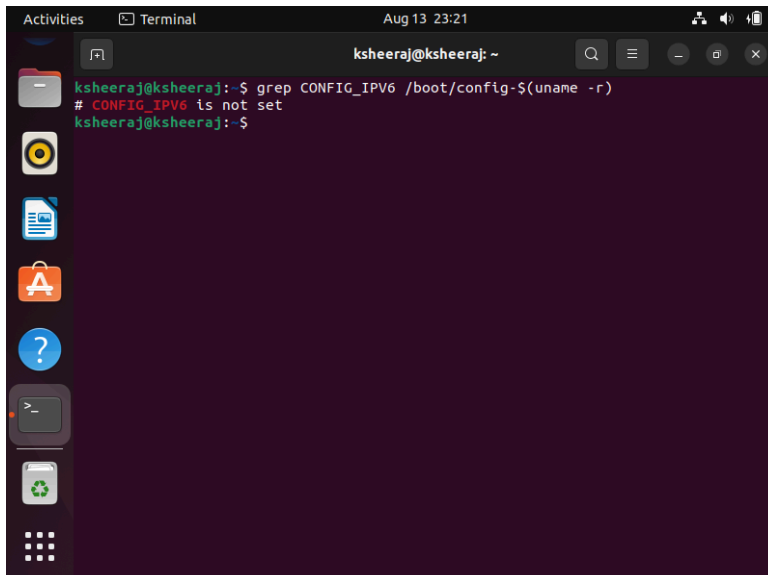
Before Installation of kernel:



A terminal window titled 'Terminal' with the date 'Aug 13 23:20' and the user 'ksheeraj@ksheeraj: ~'. The command executed is `grep CONFIG_IPV6 /boot/config-$(uname -r)`. The output lists various IPv6 configuration options, most of which are set to 'y' (yes), indicating that IPv6 is enabled in the current kernel configuration.

```
ksheeraj@ksheeraj:~$ grep CONFIG_IPV6 /boot/config-$(uname -r)
CONFIG_IPV6=y
CONFIG_IPV6_ROUTER_PREF=y
CONFIG_IPV6_ROUTE_INFO=y
# CONFIG_IPV6_OPTIMISTIC_DAD is not set
CONFIG_IPV6_NIP6=m
CONFIG_IPV6_ILA=m
CONFIG_IPV6_VTI=m
CONFIG_IPV6_SIT=m
CONFIG_IPV6_SIT_6RD=y
CONFIG_IPV6_NDISC_NODETYPE=y
CONFIG_IPV6_TUNNEL=m
CONFIG_IPV6_GRE=m
CONFIG_IPV6_FOU=m
CONFIG_IPV6_FOU_TUNNEL=m
CONFIG_IPV6_MULTIPLE_TABLES=y
CONFIG_IPV6_SUBTREES=y
CONFIG_IPV6_MROUTE=y
CONFIG_IPV6_MROUTE_MULTIPLE_TABLES=y
CONFIG_IPV6_PIMSM_V2=y
CONFIG_IPV6_SEG6_LWTUNNEL=y
CONFIG_IPV6_SEG6_HMAC=y
CONFIG_IPV6_SEG6_BPF=y
# CONFIG_IPV6_RPL_LWTUNNEL is not set
CONFIG_IPV6_IQAM6_LWTUNNEL=y
ksheeraj@ksheeraj:~$
```

After Installation of Kernel:

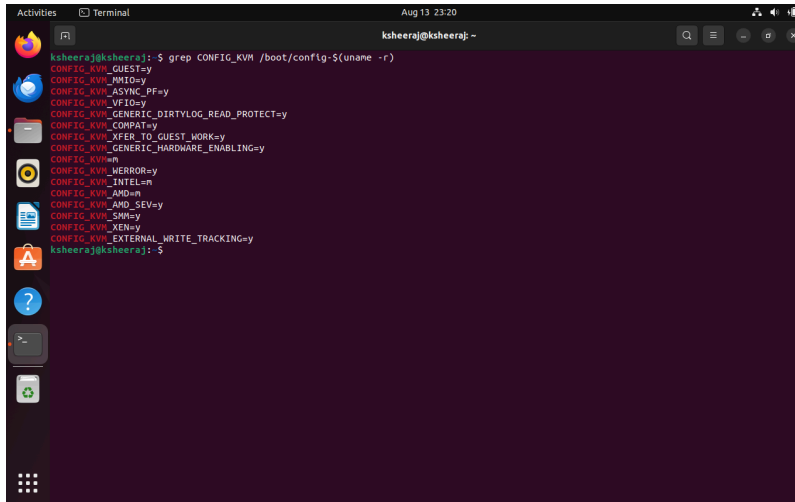


A terminal window titled 'Terminal' with the date 'Aug 13 23:21' and the user 'ksheeraj@ksheeraj: ~'. The command executed is `grep CONFIG_IPV6 /boot/config-$(uname -r)`. The output shows that `CONFIG_IPV6` is not set, indicating that IPv6 is disabled in the new kernel configuration.

```
ksheeraj@ksheeraj:~$ grep CONFIG_IPV6 /boot/config-$(uname -r)
# CONFIG_IPV6 is not set
ksheeraj@ksheeraj:~$
```

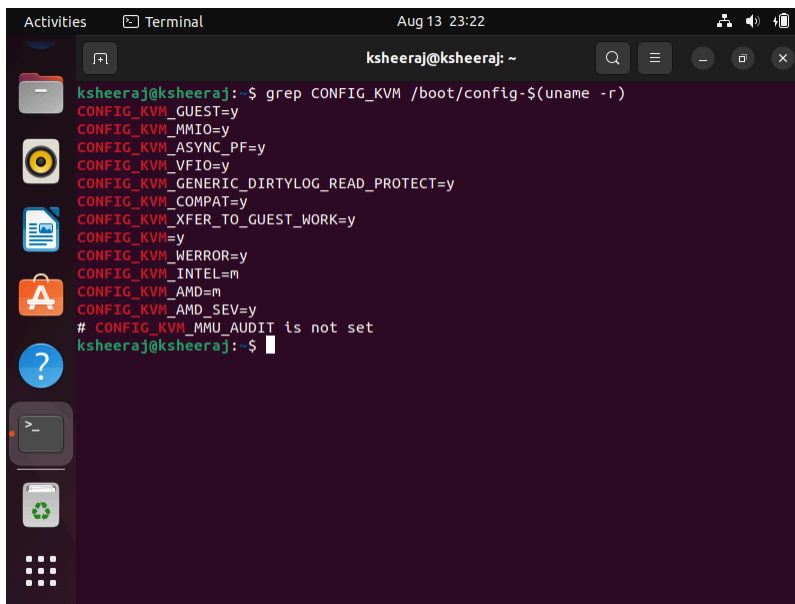
KVM (Kernel Virtual Machine)

Before Installation of kernel

A terminal window titled 'Terminal' with the date 'Aug 13 23:20' and the user 'ksheeraj@ksheeraj: ~'. The command executed is 'grep CONFIG_KVM /boot/config-\$(uname -r)'. The output lists various KVM configuration options: CONFIG_KVM_GUEST=y, CONFIG_KVM_MMIO=y, CONFIG_KVM_ASYNC_PF=y, CONFIG_KVM_VFIO=y, CONFIG_KVM_GENERIC_DIRTYLOG_READ_PROTECT=y, CONFIG_KVM_COMPAT=y, CONFIG_KVM_XFER_TO_GUEST_WORK=y, CONFIG_KVM_GENERIC_HARDWARE_ENABLING=y, CONFIG_KVM=m, CONFIG_KVM_WERROR=y, CONFIG_KVM_INTEL=m, CONFIG_KVM_AMD=m, CONFIG_KVM_AMD_SEV=y, CONFIG_KVM_SVM=y, CONFIG_KVM_XEN=y, and CONFIG_KVM_EXTERNAL_WRITE_TRACKING=y.

```
ksheeraj@ksheeraj:~$ grep CONFIG_KVM /boot/config-$(uname -r)
CONFIG_KVM_GUEST=y
CONFIG_KVM_MMIO=y
CONFIG_KVM_ASYNC_PF=y
CONFIG_KVM_VFIO=y
CONFIG_KVM_GENERIC_DIRTYLOG_READ_PROTECT=y
CONFIG_KVM_COMPAT=y
CONFIG_KVM_XFER_TO_GUEST_WORK=y
CONFIG_KVM_GENERIC_HARDWARE_ENABLING=y
CONFIG_KVM=m
CONFIG_KVM_WERROR=y
CONFIG_KVM_INTEL=m
CONFIG_KVM_AMD=m
CONFIG_KVM_AMD_SEV=y
CONFIG_KVM_SVM=y
CONFIG_KVM_XEN=y
CONFIG_KVM_EXTERNAL_WRITE_TRACKING=y
ksheeraj@ksheeraj:~$
```

After Installation of kernel

A terminal window titled 'Terminal' with the date 'Aug 13 23:22' and the user 'ksheeraj@ksheeraj: ~'. The command executed is 'grep CONFIG_KVM /boot/config-\$(uname -r)'. The output lists various KVM configuration options: CONFIG_KVM_GUEST=y, CONFIG_KVM_MMIO=y, CONFIG_KVM_ASYNC_PF=y, CONFIG_KVM_VFIO=y, CONFIG_KVM_GENERIC_DIRTYLOG_READ_PROTECT=y, CONFIG_KVM_COMPAT=y, CONFIG_KVM_XFER_TO_GUEST_WORK=y, CONFIG_KVM=y, CONFIG_KVM_WERROR=y, CONFIG_KVM_INTEL=m, CONFIG_KVM_AMD=m, CONFIG_KVM_AMD_SEV=y, and a comment '# CONFIG_KVM_MMU_AUDIT is not set'.

```
ksheeraj@ksheeraj:~$ grep CONFIG_KVM /boot/config-$(uname -r)
CONFIG_KVM_GUEST=y
CONFIG_KVM_MMIO=y
CONFIG_KVM_ASYNC_PF=y
CONFIG_KVM_VFIO=y
CONFIG_KVM_GENERIC_DIRTYLOG_READ_PROTECT=y
CONFIG_KVM_COMPAT=y
CONFIG_KVM_XFER_TO_GUEST_WORK=y
CONFIG_KVM=y
CONFIG_KVM_WERROR=y
CONFIG_KVM_INTEL=m
CONFIG_KVM_AMD=m
CONFIG_KVM_AMD_SEV=y
# CONFIG_KVM_MMU_AUDIT is not set
ksheeraj@ksheeraj:~$
```

Justification for some changes done:

1. System trusted keys and revocation keys are completely set to none

CONFIG_SYSTEM_TRUSTED_KEYS:

- This option specifies the file that contains keys to be added to the system keyring at boot. These keys are used to verify the signatures of kernel modules or firmware to ensure they are from trusted sources.
- Setting this option to **none** means that no keys are added to the system keyring. As a result, the kernel will not automatically trust any external modules or firmware unless explicitly configured to do so later.
- We used this to ensure that the system faces no security checks as we are customizing our kernel

CONFIG_SYSTEM_REVOCATION_KEYS:

- This option specifies the file that contains keys that have been revoked and should no longer be trusted by the kernel.
- Setting this to **none** means no revocation keys are provided, so the kernel will not revoke trust in any keys by default.
- As mentioned earlier too we used this to ensure that the system faces no security checks as we are customizing our kernel

2. We also changed grub timeout so as to ensure that we get to choose our own kernel while start or booting.